Process Simulation Assignment 2

Optical Lithography: From Mask Pattern to Printed Resist Feature

F437 Semiconductor Fabrication Technology, BITS Pilani K K Birla Goa Campus | **Group 7, Case C:** λ = 0.248 µm, NA = 0.55, D_{clear} = 50 mJ cm^{-2} | Ridhi (2024AAPS0032G), Rishika (2024AAPS0779G)

A one-dimensional, coherent, ideal-hard-pupil simulator was written in Python (NumPy, Matplotlib) for a 50 % duty-cycle binary mask: c_m → pupil cutoff → complex field A → $I = |A|^2$ → I_N (fixed best-focus peak) → dose $D = D_0 I_N$ → resist threshold. Every value and figure below is reproduced by the submitted notebook (2001 x-points per pitch). Lengths are in µm and dose in mJ cm^{-2}.

Task 1 Prediction before coding

R = $k_1\lambda/NA$ = 0.60 × 0.248/0.55 = **0.271 µm**; **DOF** = $k_2\lambda/NA^2$ = 0.50 × 0.248/0.55^2 = **0.410 µm** (λ/NA = 0.451 µm). With f = 0.5: c_0 = f = **0.5000**, c_1 = sin(π/2)/π = **0.3183**, c_2 = sin(π)/2π = **0**, c_3 = sin(3π/2)/3π = **−0.1061**. Also $c_{-m} = c_m$, and every even order vanishes.

Pitch	p (µm)	NA·p/λ	$m_{max} = \lfloor NA \cdot p/\lambda \rfloor$	Accepted orders
0.8 λ/NA	0.361	0.8	0	m = 0
1.2 λ/NA	0.541	1.2	1	m = 0, ±1
3.5 λ/NA	1.578	3.5	3	m = 0, ±1, ±2, ±3 ($c_{±2}$ = 0, so effectively 0, ±1, ±3)

Pitch	Predicted image	Physical reason (one sentence)
0.8 λ/NA	Uniform	p = 0.361 µm < λ/NA = 0.451 µm, so the ±1 orders fall outside the pupil and the lone zero order has nothing to interfere with.
1.2 λ/NA	Approximately sinusoidal	λ/NA < p < 2λ/NA admits only m = 0, ±1, whose interference is a single cosine on a constant background.
3.5 λ/NA	Sharper-edged	m = 0, ±1, ±3 are accepted (c_2 = 0), and the ±3 pair adds a higher spatial frequency that steepens the edges.

These predictions were recorded before the simulator was written and are not revised.

Task 2 Aerial-image simulator (order capture)

Implemented: $m_{max} = \lfloor NA\ p/\lambda \rfloor$; $A(x,\Delta z) = \Sigma\ c_m\ e^{i2\pi mx/p}\ e^{-in\lambda\Delta z(m/p)^2}$ over accepted orders (fields added before squaring); $I = |A|^2$; $I_N = I/\max_x I(x,0)$, with the same best-focus peak at every focus; $M = (I_{clear} - I_{dark})/(I_{clear} + I_{dark})$, $I_{clear} = I_N(0)$, $I_{dark} = I_N(p/2)$.

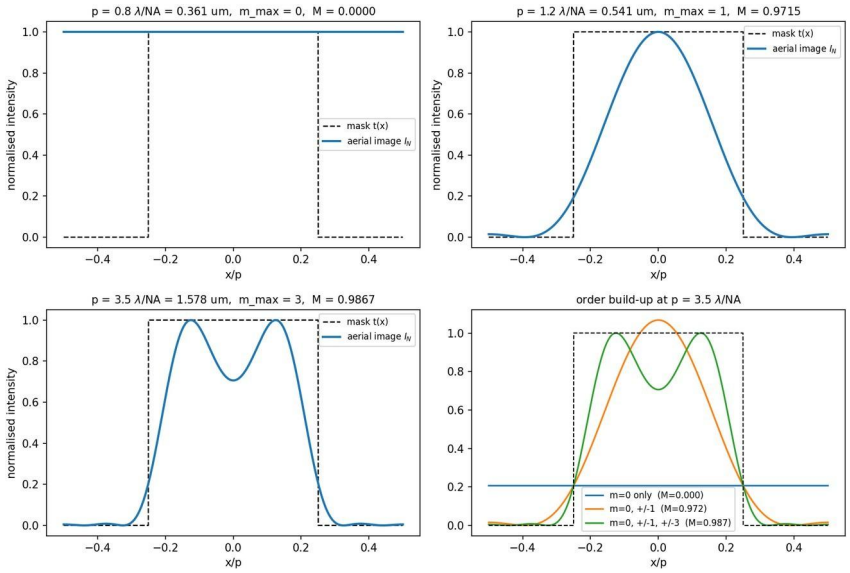


Figure 1. Ideal mask transmission and best-focus aerial image against x/p for p = 0.8, 1.2, 3.5 λ/NA, with m_{max} and M stated on each panel. The fourth panel builds up the 3.5 λ/NA image one order group at a time (M = 0, 0.972, 0.987).

Prediction vs simulation. The 0.8 and 1.2 λ/NA predictions agree (flat image, I_N = 1, M = 0; cosine-like fringe, M = 0.9715). At 3.5 λ/NA the edges are steeper as predicted (M = 0.9867), but the prediction was incomplete: the top flattens into a double peak, with $I_N(0)$ = 0.706 below the true peak of 1.0, so $I_{clear} = I_N(0)$ understates the peak.

- **Zero order (m = 0):** carries no spatial information. It sets the average intensity c_0^2; alone it gives a flat image (I_N = 0.2065 everywhere at 3.5 λ/NA, M = 0).
- **First-order pair (m = ±1):** interferes with m = 0 to form the fundamental cosine, $A = c_0 + 2c_1\cos(2\pi x/p)$. It creates almost all the contrast (M: 0 → 0.9715) but gives a rounded profile that overshoots (peak I_N = 1.067).
- **Third-order pair (m = ±3):** c_3 = −0.106 is negative, so it subtracts at the clear centre (1.067 → 0.706) and cancels residual field in the dark region (A(p/2) from −0.137 to +0.076). The edges steepen and M rises only slightly (0.9715 → 0.9867).

Verification (2.4). $\max|I(x) - I(-x)| = 1.2\times10^{-15}$ (symmetric about x = 0); $\min I_N = 2.0\times10^{-7} \geq 0$; with only m = 0 (p = 0.8 λ/NA) I_N = 1 to machine precision and M = 0.

Task 3 Lithographic transfer curve

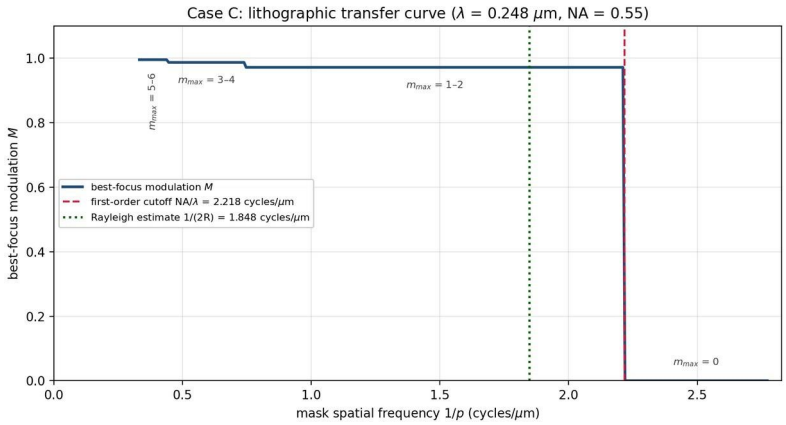


Figure 2. Best-focus modulation M against mask spatial frequency 1/p (301 values from 0.15 to 1.25 NA/λ) for the ideal hard pupil, with the first-order cutoff NA/λ = 2.218 cycles/μm and the Rayleigh estimate 1/(2R) = 1.848 cycles/μm marked.

Steps and plateaus. M depends only on which orders are accepted, $m_{max} = \lfloor NA/(\lambda \cdot (1/p)) \rfloor$. Because $c_2 = c_4 = \ldots = 0$, M changes only when an **odd** order enters or leaves the pupil: $m_{max} = 1, 2$ give M = 0.9715, $m_{max} = 3, 4$ give 0.9867 and $m_{max} = 5, 6$ give 0.9952, with steps at 1/p = NA/3λ = 0.739 and NA/5λ = 0.444 cycles/μm. The abrupt drop to M = 0 at NA/λ = 2.218 cycles/μm occurs because the hard pupil rejects m = ±1 completely, leaving only the flat zero order.

Rayleigh comparison. R = k_1λ/NA corresponds to p = 2R, i.e. 1/p = 1.848 cycles/μm (0.833 NA/λ), where the simulation still gives M = 0.9715: the ideal pupil passes finer pitches than Rayleigh predicts. The ideal cutoff p = λ/NA corresponds to $k_1 = 0.5$, whereas Rayleigh uses $k_1 = 0.6$ as a conservative practical margin. Both scale as λ/NA, so they are related, but they are not identical: the cutoff is a sharp geometric condition on the first order, while Rayleigh is an empirical screening rule.

Pattern modulation vs MTF. This curve is the best-focus contrast of one specific pattern (a 50 % duty-cycle binary grating under coherent light), and it falls abruptly because the pupil is an ideal hard cutoff. A complete experimental lens MTF describes contrast transfer for any spatial frequency and rolls off gradually, because it includes partial coherence, aberrations, defocus, flare and detector or resist response.

Task 4 Focus and numerical aperture

Working pitch p = 1.5 λ/NA = 0.6764 μm, held fixed. DOF = 0.4099 μm at NA = 0.55 and 0.2847 μm at NA = 0.66 (+20 %).

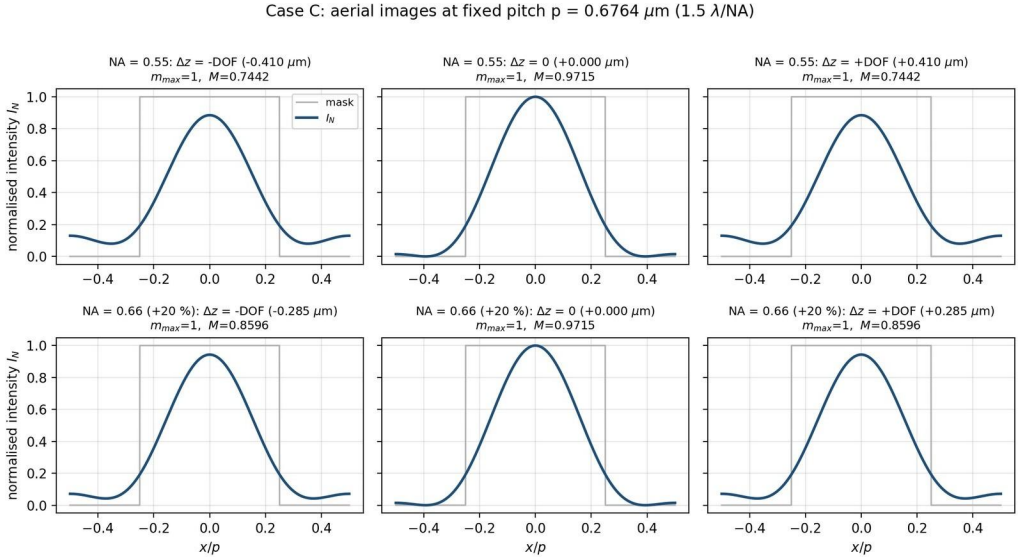


Figure 3(a). Aerial images at −DOF, 0 and +DOF for NA = 0.55 (upper row) and NA = 0.66 (lower row), with M on each panel.

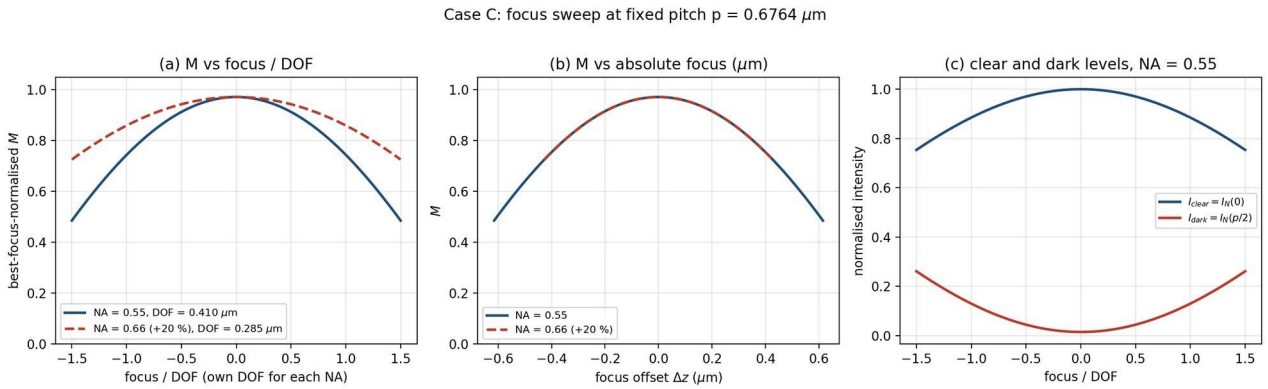


Figure 3(b). M against focus/DOF (each NA with its own DOF), M against absolute focus, and the clear and dark levels at NA = 0.55.

	DOF (μm)	m_{max}	M at best focus	M at ±DOF	M at ±1.5 DOF	R (μm)	Cutoff pitch λ/NA (μm)
NA = 0.55	0.4099	1	0.9715	0.7442	0.4858	0.2705	0.4509
NA = 0.66 (+20 %)	0.2847	1	0.9715	0.8596	0.7257	0.2255	0.3758

Symmetry. M(−DOF) = M(+DOF) = 0.7442 and the images are identical. The defocus phase $-\pi\lambda\Delta z(m/p)^2$ is odd in Δz and the c_m are real and even in m, so A(x,−Δz) = A*(x,Δz) and I is even in Δz. For m_{max} = 1, M = $M_0\cos\varphi$ with $\varphi = \pi\lambda\Delta z/p^2 = 40°$ at one DOF. Asymmetry would need aberrations (e.g. spherical), which are not modelled. M falls to 0.4858 at ±1.5 DOF because I_{dark} rises (0.014 → 0.130 at DOF) while I_{clear} falls (1.000 → 0.885).

+20 % NA. At this fixed pitch NA·p/λ goes from 1.5 to 1.8, so m_{max} stays 1 and the best-focus image and M are unchanged. The defocus factor contains no NA, so M against absolute focus is identical for both NA; the NA = 0.66 curve looks flatter against focus/DOF only because its DOF is shorter (0.2847 μm).

Trade-off. *Improves:* R falls from 0.2705 to 0.2255 μm and the cutoff pitch from 0.451 to 0.376 μm, so finer pitches (more diffraction orders) can be printed. *More focus-sensitive:* DOF ∝ $1/NA^2$ shrinks by 1/1.44 = 0.69, so the focus tolerance in μm tightens. *Immersion:* NA = n sin θ with sin θ ≤ 1; in air n ≈ 1, so NA < 1, whereas a liquid between lens and wafer (water, n ≈ 1.44 at 193 nm) raises the limit and allows NA > 1.

Task 5 From dose to printed CD

At p = 0.6764 μm the target opening is CD_t = p/2 = 0.3382 μm. I_N(p/4, 0) = 0.19351, so **$D_{target} = D_{clear}/I_N(p/4,0) = 50/0.19351 = 258.38$ mJ cm^{-2}**. CD is taken from linearly interpolated threshold crossings of D(x) = $D_0 I_N(x) \geq D_{clear}$; the same D_{target} is used at +0.75 DOF.

Dose / D_{target}	Dose (mJ cm^{-2})	CD, best focus (μm)	CD / CD_t	CD, +0.75 DOF (μm)	CD / CD_t
0.8	206.70	0.3182	0.9409	0.3155	0.9329
1.0	258.38	0.3382	1.0000	0.3382	1.0000
1.2	310.06	0.3529	1.0436	0.3555	1.0512

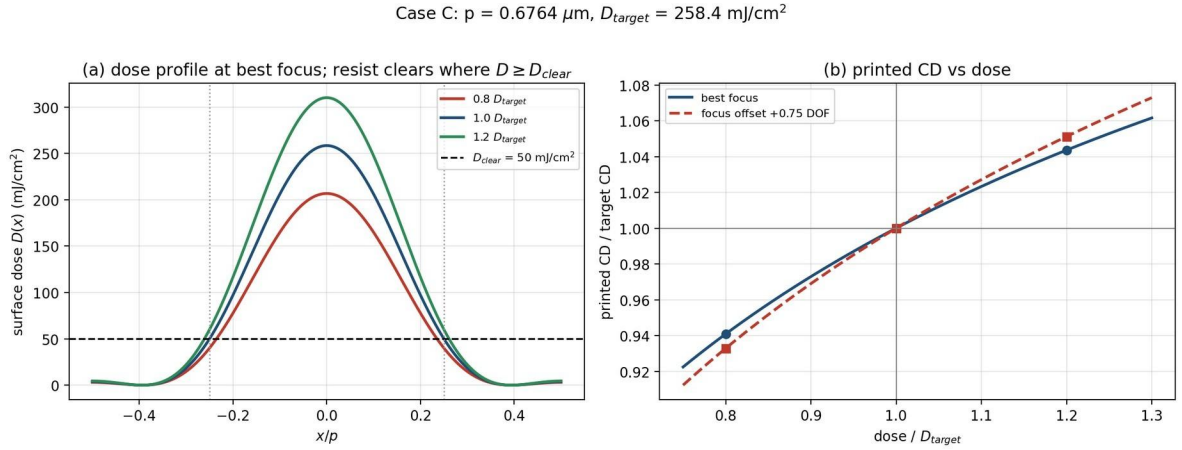


Figure 4. (a) Best-focus dose profile at 0.8, 1.0 and 1.2 D_{target} against D_{clear}. (b) Printed CD/target CD against dose/D_{target} from 0.75 to 1.30, at best focus and at +0.75 DOF.

Why dose widens the opening. D(x) = $D_0 I_N(x)$ is highest at the clear centre and falls toward the dark centre. Raising D_0 scales the profile up while D_{clear} stays fixed, so the crossing D(x) = D_{clear} moves outward to a lower-intensity position and the cleared (exposed) width grows. Near target, +1 % dose gives about +0.25 % CD at best focus and +0.29 % at +0.75 DOF, since defocus flattens the edge slope. The curves meet at 1.0 because for m_{max} = 1 the field at x = p/4 equals c_0 at every focus, so I_N(p/4) does not change with focus; defocus matters only when dose is off target.

Zero order only (p = 0.8 λ/NA). The image is flat (I_N = 1), so D(x) is the same everywhere. Below D_{clear} nothing clears (CD = 0); at or above it the whole pitch clears (CD = p = 0.3607 µm, i.e. the line disappears). Over 0.75 to 1.30 D_{target} only these two values occur. Dose scales intensity but cannot create contrast, so recovering the pattern needs at least the first order (larger NA, shorter λ or larger pitch).

Task 6 Verification, limitations and viva

Grid convergence. CD at 1.2 D_{target}, best focus (1.2× is used because at exactly D_{target} the threshold sits on the p/4 grid point and CD = p/2 on any grid):

x-points N	CD (µm)	Change vs coarser grid	Deviation from N = 4001
1001	0.35292663	–	$+1.3\times10^{-4}$ %
2001	0.35292632	-8.8×10^{-5} %	$+4.4\times10^{-5}$ %
4001	0.35292617	-4.4×10^{-5} %	0

The 2001→4001 change (4.4×10^{-5} %) is far below a 0.01 % criterion, so **N = 2001** (dx = 0.34 nm, also the assignment minimum) is retained.

Limitations. (1) *Partial coherence:* one on-axis point source; a real source of finite extent σ changes contrast, the accepted orders and the cutoff frequency. (2) *Resist physics:* a single surface-dose threshold ignores absorption through depth, standing waves, post-exposure-bake diffusion and development rate, so sidewalls and true CD are not predicted. (3) *One-dimensional periodic geometry:* only an infinite line-and-space grating is modelled, with no isolated features, 2-D corners, line ends or proximity effects. (4) *Ideal lens and mask:* a hard pupil (no roll-off, aberrations or flare), a thin binary mask (no 3-D mask or polarization effects) and scalar paraxial defocus.

Partial coherence, conceptually. Each point of an extended source illuminates the mask as a tilted plane wave, which shifts the diffraction orders relative to the pupil. For one source point the accepted orders are added as complex fields, exactly as in this simulator, giving a coherent intensity $|A_s(x)|^2$. Different source points are mutually incoherent, so their intensities, not their fields, are averaged over the source: $I(x) = \int S(s)|A_s(x)|^2\, ds$.

Verification ledger (condensed from the 19 checks printed by the notebook; all PASS):

Equation or output checked	Method of checking	Expected behaviour	Outcome
$I_N(x) = I_N(-x)$, best focus, 3 pitches	$\max\|I(x) - I(-x)\|$	≈ 0	1.2×10^{-15}, PASS
Non-negative intensity	min I_N over 3 pitches	≥ 0	2.0×10^{-7}, PASS
Only m = 0 accepted (0.8 λ/NA)	$\max\|I_N - 1\|$ and M	I_N = 1, M = 0	0 and 0, PASS
Peak normalisation	max I_N at best focus	= 1	1.000000, PASS
Even c_m vanish; $\Sigma c_m^{\,2} = f$	$\max\|c_2 \ldots c_{10}\|$; sum to $\|m\| \le 20001$	0; 0.5	1.9×10^{-17}; 0.49999, PASS
Talbot self-imaging (m ≤ 1)	$\max\|I(\Delta z = 2p^2/\lambda) - I(0)\|$	≈ 0 (phase 2π)	5.6×10^{-16}, PASS
M = 0 above cutoff; M constant on plateaus	max|M| for 1/p > NA/λ; spread within groups	0; 0	0; 4 plateaus, PASS
Focus symmetry $I(+\Delta z) = I(-\Delta z)$	$\max\|I_N(+DOF) - I_N(-DOF)\|$	≈ 0 (conjugate field)	5.6×10^{-16}, PASS
$M(focus) = M_0 \cos\varphi$ for $m_{max} = 1$	$\max\|M_{num} - M_{analytic}\|$	agree to ≈ 10^{-9}	3.3×10^{-16}, PASS
DOF ∝ $1/NA^2$; best-focus M same for both NA	DOF ratio; M(0) at each NA	0.6944; equal	0.6944; 0.9715 both, PASS
$CD(D_{target}) = p/2$, independent of focus	CD vs p/2 at best focus and +0.75 DOF	p/2 = 0.338182 µm	0.338182 µm both, PASS
CD non-decreasing with dose (positive resist)	np.diff of both CD curves	monotonic increase	PASS
m = 0 only: CD takes only 0 or p	CD over 0.75 to 1.30 D_{target}	no intermediate CD	{0, 0.3607} µm, PASS
Grid convergence of CD	N = 1001, 2001, 4001 at 1.2 D_{target}	change < 0.01 %	-4.4×10^{-5} %, PASS

Contribution statement and AI Use Log (excluded from the five-page limit)

Group 7, Case C. Each member individually explains one equation, one code block and one physical result in the viva.

Member	Implemented	Checked
Ridhi (2024AAPS0032G)	Fourier coefficients, pupil cutoff, complex field, normalised intensity and modulation (Task 2); transfer-curve sweep (Task 3); focus and NA analysis (Task 4).	Symmetry about x = 0, non-negative intensity, m = 0-only image, Talbot self-imaging; M = $M_0\cos\varphi$ against the numerical focus sweep; ±DOF symmetry.
Rishika (2024AAPS0779G)	Dose and threshold model (cd_at(), printed_cd()) and the dose-to-CD curves (Task 5); grid-convergence study and verification ledger (Task 6).	$CD(D_{target}) = p/2$; CD non-decreasing with dose; m = 0-only CD limited to {0, p}; grid convergence at 1001, 2001 and 4001 points.

AI Use Log

AI was used to write the code and draw inferences from the graphs. Task 1 calculations were done by hand as well as the predictions, however the markdown code was written by AI.